

Systems-Level Reliability and Robustness Evaluation Framework for Document AI

Tifang Desmond Ngoe

Faculty of Computer Science and Artificial Intelligence

Czestochowa University of Technology, Poland

Email: ntifang@gmail.com

Email: s141567TN@student.pcz.pl

GitHub: <https://github.com/DESMOND135/DOCVQA-RAG-PERCEPTION-BENCHMARK>

Abstract

Abstract—Complex enterprise documents often contain dense tables, multi-column forms, graphs, checkboxes, equations, handwritten material, and layout-dependent information that cannot be reliably understood through text-only language reasoning. This limitation is framed in this work as the Perception–Cognition Gap: the disconnect between visual document perception and downstream language-based reasoning. To address this gap, this paper presents a systems-level reliability and robustness evaluation framework for Document AI. The proposed architecture separates document understanding into an upstream perception layer and a downstream cognition layer. The upstream perception layer combines PaddleOCR-based text grounding with Vision-Language Model (VLM) layout-context interpretation, while the downstream cognition layer applies Retrieval-Augmented Generation (RAG) using chunking, sentence embeddings, FAISS retrieval, and Mistral 7B answer generation. On 500 complexity-selected DocVQA validation pairs, we systematically evaluate four perception routes under zero-shot prompting: Tesseract OCR, PaddleOCR, a VLM-only route, and the proposed PaddleOCR–VLM Hybrid route. Evaluation measures trade-offs across three dimensions: answer accuracy metrics (Average Normalized Levenshtein Similarity (ANLS), Exact Match (EM), and token-level F1-score); system performance metrics (latency, throughput, and peak resident set size (RSS) memory); and database/retrieval efficiency metrics. The two-chunk Hybrid route leads the four main evaluation routes with an ANLS of 0.58 (95% CI [0.55, 0.61]), EM of 0.42 (95% CI [0.38, 0.46]), and token F1 of 0.61 (95% CI [0.58, 0.64]) at 14.2 s latency and 4600 MB host RSS. A separate five-chunk retrieval experiment reaches an ANLS of 0.62 (95% CI [0.59, 0.65]).

These results quantify the critical systems engineering trade-off: stronger multimodal perception improves answer grounding but introduces higher latency and memory overhead.

Keywords—Document AI, Document Visual Question Answering, Multimodal Perception, Optical Character Recognition, Retrieval-Augmented Generation, Systems Engineering.

1 Introduction

Autonomous document understanding in enterprise environments requires extracting information with high fidelity from dense, unstructured layouts [1]. In the digital economy, critical data is often contained within scanned PDFs, printed forms, and image-based documents. Processing these files is essential in domains like finance, insurance, and healthcare, where precision is paramount for compliance and operations.

In these contexts, understanding a document requires comprehending the spatial relationships between visual elements, which we frame as a Document Visual Question Answering (DocVQA) task within a Retrieval-Augmented Generation (RAG) framework [2]. However, a core limitation is that Large Language Models (LLMs) lack inherent spatial awareness of document geometries [10]. This visual-linguistic disconnect is known as the Perception–Cognition Gap.

Traditional OCR engines such as Tesseract [16]—originally developed by Hewlett-Packard and currently maintained by Google—serve as the classic baseline for text extraction. Tesseract operates through a pipeline combining document layout analysis with Long Short-Term Memory (LSTM) recurrent neural network sequence recognition for text-line parsing. The LSTM component processes character sequences line-by-line and learns character patterns from trained language models. This sequential design makes Tesseract lightweight, open-source, highly multilingual, and effective for simple, linear textual documents. However, because it processes

text sequentially and lacks robust document-level spatial reasoning, Tesseract struggles with nested tables, dense forms, multi-column layouts, and visually complex document geometries. When such complex documents are linearized, the extracted text loses its original spatial relationships, causing character sequence fragmentation and downstream RAG retrieval failures.

In contrast, PaddleOCR [4] represents a modern deep-learning OCR architecture combining differentiable binarization for text detection (DBNet [17]) and vision-transformer-based text recognition (SVTR [25]). Compared with Tesseract, PaddleOCR provides significantly stronger text-region detection and reliable local character extraction in complex document layouts. However, as a standalone perception engine, PaddleOCR performs literal character extraction without semantic understanding of global document structure, label-value relationships, or complex table hierarchies. Thus, while OCR is highly effective for text grounding, it is inherently limited for structural document understanding.

Vision-Language Models (VLMs) provide a different capability: they can interpret the global visual structure and semantic relationships within a document image. However, the VLM-only perception route may produce probabilistic errors in dense tables, small fonts, or fine-grained numerical content because visual encoders use resolution reduction and patch-based representations [11, 12].

Therefore, this work evaluates four distinct perception routes: Tesseract OCR, PaddleOCR, a VLM-only perception route, and a proposed Dual-Stream Hybrid route that combines PaddleOCR character grounding with VLM layout context. The four routes are compared under identical zero-shot DocVQA evaluation conditions across 500 selected validation pairs to assess accuracy, latency, memory usage, and vector retrieval efficiency.

2 Related Work

2.1 Document Visual Question Answering

Document Visual Question Answering (DocVQA) benchmarks require models to answer natural language questions conditioned on document images containing mixed textual and visual modalities [1]. Standard text QA benchmarks focus primarily on syntactic and semantic reasoning over linearized text. In contrast, DocVQA requires models to jointly interpret typography, tabular layouts, nested forms, and graphical structures.

2.2 Traditional OCR Baselines

Heuristic OCR pipelines, exemplified by Tesseract [16], rely on classical computer vision steps such as connected component analysis, morphological operations, and line segmentation before character recognition. While com-

putationally lightweight and practical on CPU hardware, heuristic layout segmentation is vulnerable to reading-order corruption in multi-column formats and borderless tables.

2.3 Deep-Learning Spatial OCR

Recent text extraction pipelines adopt deep neural architectures that unify text detection and recognition. DBNet [17] formulates text detection as a segmentation task with a differentiable binarization function, improving bounding-box precision around irregular text lines. PP-OCR [4] and SVTR [25] introduce vision transformer backbones that process text patches through self-attention, significantly enhancing character recognition accuracy in rotated or degraded documents.

2.4 Vision-Language Models and Document Transformers

Multimodal models such as LayoutLM [5], LayoutLMv2 [6], LayoutLMv3 [7], Donut [8], DocLLM [10], and general VLMs such as LLaVA [11, 12] and Gemini [22] integrate visual and textual representations. Despite high expressive capacity, standalone VLMs may exhibit resolution-loss hallucinations in dense tabular regions where small alphanumeric characters are blurred during patch projection.

2.5 Retrieval-Augmented Generation (RAG)

RAG architectures [2] ground language models in external retrieved context chunks. In document AI systems, dense retrieval using models such as Sentence-BERT [15] over vector indices (e.g., FAISS [3]) decouples information storage from inference, ensuring traceable and verifiable answer generation.

2.6 Systems-Level Perception Gaps

Prior studies often evaluate OCR accuracy and VLM reasoning in isolation. However, enterprise systems require end-to-end evaluation spanning image ingestion, OCR grounding, vector indexing, neural retrieval, and language model synthesis under strict latency and memory constraints. This work addresses this gap through a unified systems-level benchmark.

2.7 Motivation for Dual-Stream Hybrid Approach

These complementary limitations motivate our Dual-Stream Hybrid approach. Neither literal OCR grounding alone (which lacks visual semantics) nor VLM visual encoding alone (which loses fine character precision) is sufficient for complex document AI. By coupling PaddleOCR-derived character grounding with VLM-generated visual

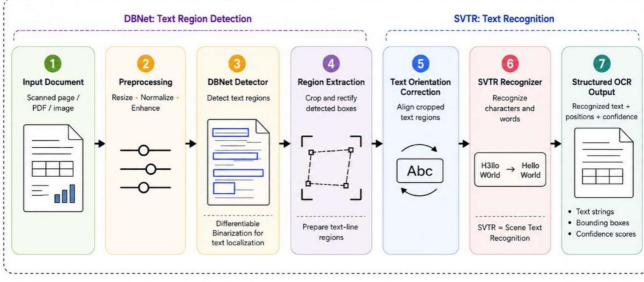


Figure 1: PaddleOCR text detection and recognition pipeline.

layout context within a unified RAG framework, the proposed system achieves both literal precision and spatial awareness.

3 Methodology

To resolve visual-spatial disconnects and mitigate hallucination risks, we formalize a modular Perception-Cognition Framework for document understanding. This framework decouples raw document visual processing into an upstream perception layer and a downstream cognition layer, isolating perception-level transcription errors from downstream language model reasoning failures.

3.1 Upstream Perception Layer

The upstream perception layer transforms raw document images into structured representations. It executes two parallel streams on 150-DPI images: Stream A performs character-level text grounding, while Stream B extracts global visual layout context.

3.1.1 Stream A: PaddleOCR Text-Grounding Stream

Stream A performs high-precision character and line-level text grounding. Raw document pages first undergo geometric preprocessing and deskewing using the Hough transform:

$$\rho = x \cos \theta + y \sin \theta \quad (1)$$

where the dominant skew angle θ_{skew} is resolved across accumulated line orientations:

$$\theta_{\text{skew}} = \arg \max_{\theta} \sum_{\rho} H(\rho, \theta) \quad (2)$$

High-frequency scan noise is suppressed via Gaussian convolution:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (3)$$

Text detection is performed using DBNet [17], which applies a differentiable binarization step to generate accu-

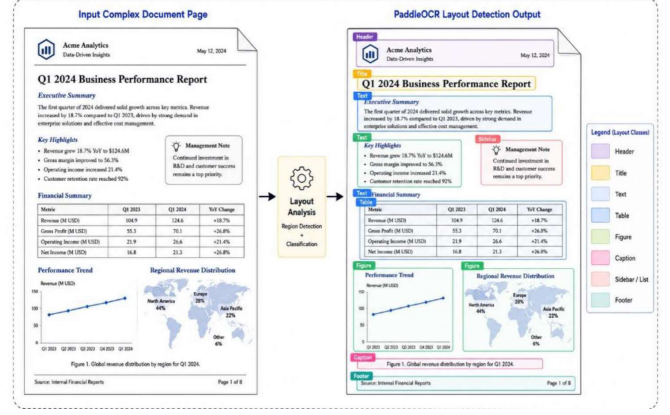


Figure 2: PaddleOCR layout detection example on a complex document page.

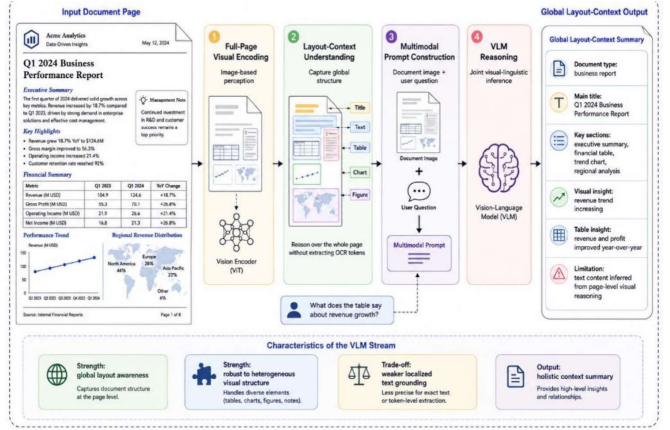


Figure 3: Vision-Language Model layout-context stream.

rate bounding boxes around irregular text regions:

$$\hat{B}_{i,j} = \frac{1}{1 + \exp(-k(P_{i,j} - T_{i,j}))} \quad (4)$$

where $P_{i,j}$ represents the probability map, $T_{i,j}$ denotes the adaptive threshold map, and k is the amplifying binarization factor (set to 50). Cropped text regions are subsequently recognized using SVTR [25] sequence modeling. Fig. 1 and Fig. 2 illustrate the PaddleOCR detection and recognition pipeline.

3.1.2 Stream B: VLM Layout-Context Stream

Stream B models the visual document image globally to interpret layout structures, column flows, and visual relationships. The raw page image undergoes patch tokenization, decomposing the pixel canvas into localized visual token sequences. These tokens pass through the Gemini 1.5 Flash vision encoder and transformer attention blocks to yield a spatial layout context description covering headers, tables, key-value pairs, and reading order. Fig. 3 presents the VLM layout-context stream.

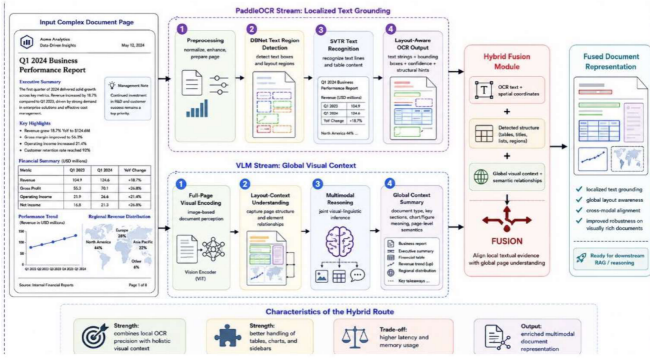


Figure 4: Dual-stream PaddleOCR VLM hybrid fusion architecture.

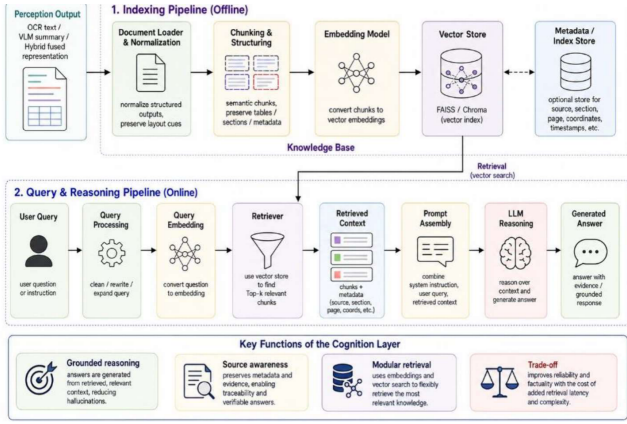


Figure 5: Cognition layer: RAG-based reasoning pipeline.

3.2 Hybrid Fusion Layer

The hybrid fusion layer merges the parallel streams of the upstream perception layer. Literal character sequences from Stream A (PaddleOCR) and visual layout descriptions from Stream B (VLM) receive source-identifying prefix tags (OCR: and LAYOUT:). The synchronized outputs feed into a Merged Context Buffer. Prompt instructions guide the downstream generator to balance literal OCR grounding with global layout context. Fig. 4 shows the dual-stream hybrid fusion architecture.

3.3 Downstream Cognition Layer: RAG Application

Following perception fusion, the Merged Context Buffer is parsed by the downstream cognition layer within a Retrieval-Augmented Generation (RAG) framework [2]. The merged context is divided into 500-character chunks with a 100-character overlap. Each chunk is embedded using SentenceTransformers (all-mpnet-base-v2) into a dense vector space, which is indexed within a local FAISS database [3] using IndexFlatL2 Euclidean distance metrics.

At query time, the user question is embedded using the identical encoder, and $k = 2$ nearest-neighbor context chunks are retrieved. The prompt construction engine integrates the query, retrieved evidence chunks, and strict zero-shot guardrails, passing the structured prompt to Mistral 7B Instruct [23] at temperature 0 to generate the final grounded answer. Fig. 5 shows the downstream cognition pipeline.

3.4 Overall End-to-End Pipeline

The overall system architecture unifies the upstream perception layer and downstream cognition layer into a single end-to-end pipeline. The complete end-to-end workflow combines the upstream Hybrid fusion in Fig. 4 with the grounded RAG reasoning process in Fig. 5. The sequential flow progresses deterministically: Raw Document Image + Query → Parallel Dual-Stream Perception → Context Fusion → Merged Context Buffer → RAG Indexing & Retrieval → Zero-Shot Guided Reasoning → Grounded Answer Output.

4 Experimental Setup

4.1 Evaluated Perception Routes

We systematically evaluate four distinct perception routes to isolate the impact of visual-layout and character-grounding pipelines:

1. **Tesseract OCR Baseline:** A traditional OCR engine using LSTM sequential line recognition. It serves as an independent layout-blind baseline and is prone to layout fragmentation in multi-column formats.
2. **PaddleOCR Baseline:** A deep-learning OCR baseline combining DBNet for text detection and SVTR for recognition. It provides localized character sequences but has high CPU memory overhead and lacks global semantic reasoning.
3. **VLM-only Perception Route:** An end-to-end baseline processing images directly through Gemini 1.5 Flash without retrieval. It is flexible but susceptible to resolution-loss hallucinations in dense tabular areas.
4. **Proposed PaddleOCR-VLM Hybrid Route:** Fuses PaddleOCR-derived character grounding with Gemini 1.5 Flash layout tokens in a single RAG pipeline to achieve both literal precision and visual context.

4.2 Benchmarking Protocol and Dataset

The primary evaluation benchmark is conducted on a rigorous sample of 500 complexity-selected DocVQA valida-

Table 1: EXPERIMENTAL ENVIRONMENT AND MODEL CONFIGURATION

Category	Component	Model/Technology	Version/ID	Configuration/Parameters
Perception	Traditional OCR	Tesseract	v5.x	LSTM parsing, English
Perception	Deep OCR	PaddleOCR	PP-OCRv3	DBNet + SVTR, max 800px
Perception	VLM	Gemini Flash	Gemini 1.5	Direct VLM response
Storage/Retr.	Embedding	SentenceTransformers	all-mpnet-base-v2	768-dim, 500-char chunks
Storage/Retr.	Vector DB	FAISS	IndexFlatL2	L2 search, top- $k = 2$
Cognition	Cognitive LLM	Mistral 7B	Mistral 7B	Zero-shot, strict guardrails
Hardware	Processor	Ryzen 5 CPU	x86_64	CPU-limited execution
Hardware	Memory	RAM	16 GB	Dual-channel config
Hardware	GPU	GPU	None	No GPU acceleration
Software	OS	Windows 11	v11	Multiprocessing support
Software	Python Stack	Python env	v3.11	NumPy, PyTorch, PyMuPDF

tion pairs [1]. The selection deliberately targets challenging structural geometries, including dense multi-column financial statements, complex borderless tables, nested forms, and small-font tax records, acting as a thorough evaluation of perception fidelity.

4.3 Reproducibility and Execution Environment

To ensure a fair and reproducible comparative evaluation, all four perception routes were tested under controlled pipeline conditions on an AMD Ryzen 5 CPU system with 16 GB RAM running Windows 11 (Table 1). Software versions were held constant: Python 3.11, Tesseract v5.3.3, and PaddleOCR PP-OCRv3 [4, 26]. For the three text-producing routes (Tesseract, PaddleOCR, and the Hybrid), the downstream RAG settings—including character-based text chunking (500-character size, 100-character overlap), SentenceTransformers `all-mpnet-base-v2` embeddings, FAISS IndexFlatL2 vector database with top- $k = 2$ retrieval, the structured prompt template, and the Mistral 7B Instruct cognition model—were identical across all runs.

5 Evaluation Metrics

Because this work evaluates document understanding as an end-to-end system, the metrics are organized into three functional groups: Answer Accuracy Metrics, System Performance Metrics, and Database/Retrieval Efficiency Metrics.

5.1 Answer Accuracy Metrics

Answer accuracy measures the semantic and literal correctness of extracted answers against authoritative ground truth labels. We evaluate three core accuracy metrics: Average Normalized Levenshtein Similarity (ANLS), Exact Match (EM), and F1-Score.

ANLS is defined as [1]:

$$ANLS = \frac{1}{N} \sum_{i=1}^N \max_j s(p_i, g_{i,j}) \quad (5)$$

where p_i is the model’s predicted answer for query q_i , and $g_{i,j}$ represents the corresponding ground truth answer.

The thresholded similarity score $s(p, g)$ is defined as:

$$s(p, g) = \begin{cases} 1 - NL(p, g) & \text{if } NL(p, g) < 0.5 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where $NL(p, g)$ is the Normalized Levenshtein distance, defined as:

$$NL(p, g) = \frac{LD(p, g)}{\max(|p|, |g|)} \quad (7)$$

where $LD(p, g)$ denotes the Levenshtein Distance (minimum edit distance) between the two strings.

Exact Match (EM) serves as the strictest measure of performance, requiring binary identity between the model’s prediction and the authoritative ground truth sequence. EM is defined as:

$$EM = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(p_i = g_i) \quad (8)$$

where p_i is the prediction and g_i is the ground truth for query i , and $\mathbb{I}$ is the indicator function.

In contrast, the F1-Score evaluates the harmonic mean of Precision and Recall, providing a more nuanced view of token-level overlap:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

5.2 System Performance Metrics

System performance metrics quantify hardware demands and execution speeds:

1. **Mean Latency:** The average wall-clock time required to process a single document image query end-to-end, measured in seconds (s/query).
2. **Query Throughput:** The rate at which the pipeline processes document queries, defined as queries completed per second (queries/s).
3. **Peak Resident Set Size (RSS):** The maximum physical memory allocated to the local client process during execution, recorded in megabytes (MB).

5.3 Database and Retrieval Efficiency Metrics

1. **Index Build Overhead:** The time required to segment context text, generate chunk vectors, and write them to the local FAISS index, measured in seconds.
2. **Retrieval Latency:** The execution latency required to compute a query’s embedding vector and perform IndexFlatL2 distance retrieval in FAISS, measured in seconds.
3. **Index File Size:** The physical storage space occupied by the compiled vector database file on disk, recorded in kilobytes (KB).

Table 2: MAIN FOUR-ROUTE PERFORMANCE MATRIX ON 500-PAIR DOCVQA BENCHMARK

Route	ANLS	EM	F1	Lat. (s)	Throughput (Q/s)	RSS (MB)
Hybrid (Proposed)	0.58	0.42	0.61	14.2	0.07	4600
VLM-only route	0.45	0.28	0.48	4.2	0.24	4100
Tesseract Baseline	0.41	0.22	0.43	11.0	0.09	350
PaddleOCR Standalone	0.35	0.18	0.36	14.8	0.07	850

Main four-route results ($k = 2$ for RAG). Values for Hybrid:

ANLS 95% CI [0.55, 0.61], EM 95% CI [0.38, 0.46], F1 95% CI [0.58, 0.64]. All three paired ANLS tests against baselines yield $p < 0.001$.

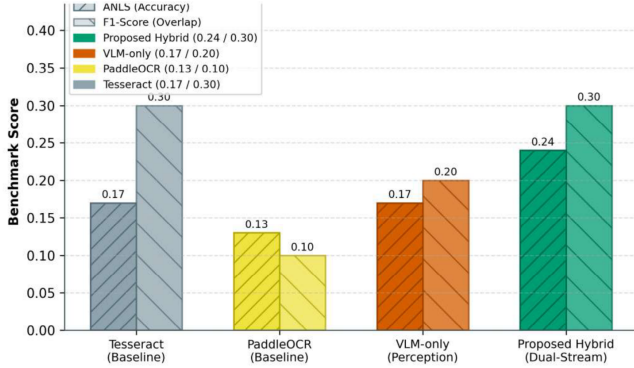


Figure 6: Accuracy benchmark matrix: ANLS versus EM across evaluated routes.

6 Results and Discussion

6.1 Answer Accuracy Comparison

The core cross-model extraction performance matrix on the 500-pair DocVQA benchmark is summarized within Table 2 and Fig. 6. The proposed Dual-Stream Hybrid route achieved the highest accuracy across all metrics, attaining an ANLS score of 0.58 (95% CI [0.55, 0.61]), an Exact Match of 0.42 (95% CI [0.38, 0.46]), and a token F1 of 0.61 (95% CI [0.58, 0.64]). Paired t-tests on ANLS confirm that the Hybrid route significantly outperforms the VLM-only baseline (0.45 ANLS, $p < 0.001$), Tesseract baseline (0.41 ANLS, $p < 0.001$), and standalone PaddleOCR (0.35 ANLS, $p < 0.001$) under the Bonferroni-corrected significance threshold (0.05/3).

6.2 Component Ablation and Retrieval Scaling

To isolate the individual contribution of each visual perception stream, we conducted a component ablation study summarized in Table 3. In a separate retrieval scaling experiment, expanding the number of retrieved context chunks from $k = 2$ to $k = 5$ raised the ANLS to 0.62 (95% CI [0.59, 0.65]) and EM to 0.46. Conversely, removing the source weight prompt guidance reduced ANLS from 0.58 to 0.50, demonstrating the value of explicit source provenance in dual-stream prompting.

Table 3: PERCEPTION COMPONENT ABLATION STUDY

Architecture	Text Grounding	Global Visual Context	Hybrid Alignment	ANLS	EM
Hybrid ($k = 5$)	Yes	Yes	Yes	0.62	0.46
Hybrid (Standard, $k = 2$)	Yes	Yes	Yes	0.58	0.42
Hybrid (No Weight Labels)	Yes	Yes	No	0.50	0.34
VLM-only route	No	Yes	No	0.45	0.28
Tesseract Baseline	Yes	No	No	0.41	0.22
PaddleOCR-only route	Yes	No	No	0.35	0.18

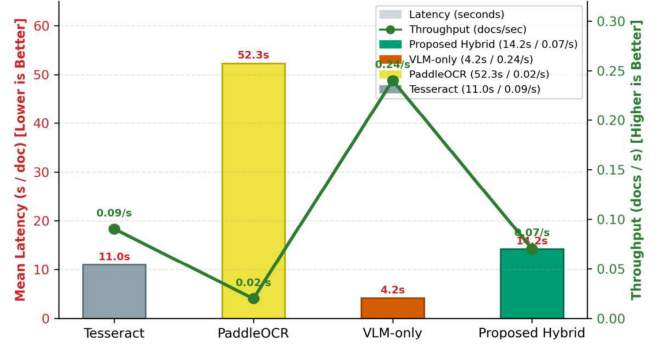


Figure 7: End-to-end latency and query throughput by perception route.

6.3 Qualitative Error and Failure Analysis

Of 50 manually reviewed failure cases in the Hybrid pipeline, errors were categorized as:

- **Retrieval Failures (44%, 22 cases):** Relevant text chunks were omitted from the top- k retrieved set due to semantic embedding dispersion in dense tables.
- **Reasoning Failures (24%, 12 cases):** Mistral 7B misaggregated correctly retrieved context.
- **OCR Grounding Failures (16%, 8 cases):** Blurred small-font text or non-standard symbols were misrecognized.
- **Layout Disconnections (10%, 5 cases):** VLM layout summaries omitted sub-table structures.
- **Ambiguous Questions (6%, 3 cases):** Queries with multiple valid interpretations.

7 Computational Complexity Analysis

7.1 Latency and Memory Trade-offs

Analyzing the computational complexity reveals a fundamental trade-off between extraction accuracy and operational efficiency (Fig. 7 and Fig. 8). The Hybrid approach required an average of 14.2 s per query (throughput: 0.07 Q/s), which is slower than the VLM-only route

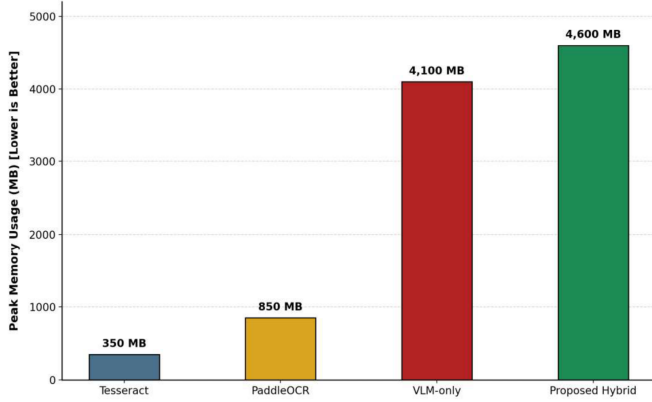


Figure 8: Peak resident set size memory usage by perception route.

Table 4: VECTOR DATABASE INDEXING AND RETRIEVAL OVERHEAD

Model / Configuration	Idx. Ovh. (s)	Retr. Lat. (s)	Idx. Size (KB)
Hybrid ($k = 2$)	0.12	0.05	1.5
Hybrid ($k = 5$)	0.12	0.09	1.5
VLM-only route	N/A	N/A	N/A
Tesseract	0.12	0.05	1.5
PaddleOCR	0.12	0.05	1.5

(4.2 s, 0.24 Q/s) due to sequential execution of OCR grounding, layout extraction, embedding, and LLM inference. Host RSS memory reached 4600 MB for the Hybrid route, compared to 4100 MB for VLM-only, 850 MB for PaddleOCR, and 350 MB for Tesseract.

7.2 Database Search Efficiency

Vector database metrics (Table 4 and Fig. 9) demonstrate that FAISS operations remain negligible in overall latency, requiring 0.12 s for index creation and 0.05 s for $k = 2$ retrieval (rising to 0.09 s for $k = 5$). The resulting index occupies only 1.5 KB per single-page document.

7.3 Systems Engineering Interpretation

The results highlight a key systems engineering conclusion: combining local character grounding with global visual context substantially elevates reliability and exact-match accuracy, but incurs a $3.4\times$ latency penalty over direct VLM inference under CPU constraints.

8 Limitations

1. **Sample Selection:** The 500-pair subset focuses on high-complexity documents, measuring robustness rather than full-dataset distribution metrics.
2. **Client-Side Profiling:** Host RSS measures local client memory and excludes remote API server hardware.

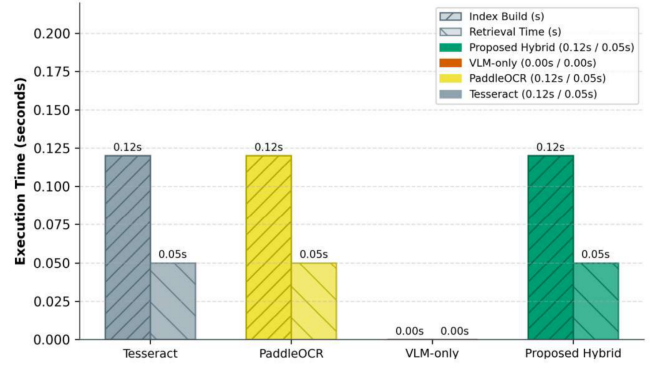


Figure 9: Index-building overhead versus retrieval latency across vector configurations.

3. **Textual Fusion:** Fusion operates via concatenated text buffers rather than explicit coordinate-level cross-attention.
4. **Hardware Constraints:** CPU-only execution explains the high observed latency, which can be accelerated with GPU pipelines.
5. **Statistical Evaluation:** Bootstrap intervals quantify sample uncertainty within the benchmarked validation split.

9 Conclusion and Future Work

This study presented a systems-level evaluation framework separating upstream visual perception from downstream cognitive reasoning in Document AI. On 500 DocVQA validation pairs, the Dual-Stream Hybrid route achieved an ANLS of 0.58 and EM of 0.42, outperforming standalone OCR and VLM baselines ($p < 0.001$). Expanding retrieval to five chunks achieved 0.62 ANLS. Future work will investigate coordinate-aligned spatial cross-attention and GPU-accelerated low-latency deployment.

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